

Section 4.1

Exercise (4.1.1). Let G act on the set A . Prove that if $a, b \in A$ and $b = g \cdot a$ for some $g \in G$, then $G_b = gG_ag^{-1}$ (G_a is the stabilizer of a). Deduce that if G acts transitively on A then the kernel of the action is $\cap_{g \in G} gG_ag^{-1}$.

Proof: Let G act on the set A . Suppose $a, b \in A$, and $b = g \cdot a$ for some $g \in G$. Let $g_1 \in G_b$ and consider $g^{-1}g_1g \cdot a$. Note that since $g \cdot a = b$ and $g_1 \in G_b$ we get that $g^{-1}g_1(g \cdot a) = g^{-1}g_1 \cdot b = g^{-1}b = a$. It follows that $g^{-1}g_1g \in G_a$, therefore $g_1 \in gG_ag^{-1}$ and thus $G_b \subseteq gG_ag^{-1}$. Let $g_1 \in gG_ag^{-1}$, therefore $g_1 = gg_ag^{-1}$ where $g_a \in G_a$. Consider $g_1 \cdot b$ and note that since $g \cdot a = b$ and $g_a \in G_a$ we get $g_1 \cdot b = gg_a(g^{-1} \cdot b) = gg_a \cdot a = g \cdot a = b$. It follows that $g_1 \in G_b$ and therefore $gG_ag^{-1} \subseteq G_b$ thus $G_b = gG_ag^{-1}$.

Suppose G acts transitively on A . Recall that the kernel of an action is the set of all $g \in G$ that map everything in A to itself, and therefore must be the intersection of all the stabilizers. Fix $a \in A$, and note that since G acts transitively on A for all $b \in A$ there exists a $g \in G$ such that $g \cdot a = b$ thus by the previous result we get, $\cap_{b \in A} G_b = \cap_{g \in G} gG_ag^{-1}$

□

Exercise (4.1.6). As in Exercise 12 of Section 2.2 let R be the set of all polynomials with integer coefficient in the independent variables x_1, x_2, x_3, x_4 and let S_4 act on R by permuting the indices of the four variables,

$$\sigma \cdot (x_1, x_2, x_3, x_4) = p(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}, x_{\sigma(4)})$$

for all $\sigma \in S_4$.

- a. Find the polynomials in the orbit of S_4 on R containing $x_1 + x_2$.

Solution: Applying the Orbit Stabilizer Theorem we know that, $|O(x_1 + x_2)| = |S_4 : \text{Stab}(x_1 + x_2)|$. Note that $\text{Stab}(x_1 + x_2) = \{1, (12), (34), (12)(34)\}$. Therefore we know that $|O(x_1 + x_2)| = 24/4 = 6$. Permuting the indices of $x_1 + x_2$ until

we have six distinct polynomials we get,

$$O(x_1 + x_2) = \{(x_1 + x_2), (x_1 + x_3), (x_1 + x_4), (x_2 + x_3), (x_2 + x_4), (x_3 + x_4)\}.$$

□

- b. Find the polynomials in the orbit of S_4 on R containing $x_1x_2 + x_3x_4$.

Solution: Applying the Orbit Stabilizer Theorem we know that, $|O(x_1x_2 + x_3x_4)| = |S_4 : \text{Stab}(x_1x_2 + x_3x_4)|$. Note that $\text{Stab}(x_1x_2 + x_3x_4) = \{1, (12), (34), (12)(34), (13)(24), (14)(23), (1423), (1324)\}$. Therefore we know that $|O(x_1x_2 + x_3x_4)| = 24/8 = 3$. Permuting the indices of $x_1x_2 + x_3x_4$ until we have 3 distinct polynomials we get,

$$O(x_1x_2 + x_3x_4) = \{(x_1x_2 + x_3x_4), (x_1x_3 + x_2x_4), (x_1x_4 + x_2x_3)\}.$$

□

- c. Find the polynomials in the orbit of s_4 on R containing $(x_1 + x_2)(x_4 + x_3)$.

Solution: Note that since the same elements are being paired together, i.e. x_1, x_2 and x_3, x_4 and since addition and multiplication are commutative we should get that $\text{Stab}(x_1x_2 + x_3x_4) = \text{Stab}((x_1 + x_2)(x_4 + x_3))$. Therefore $|O((x_1 + x_2)(x_4 + x_3))| = 24/8 = 3$. Permuting the indices of $(x_1 + x_2)(x_4 + x_3)$ until we have 3 distinct polynomials we get,

$$O((x_1 + x_2)(x_4 + x_3)) = \{(x_1 + x_2)(x_4 + x_3), (x_1 + x_3)(x_4 + x_2), (x_1 + x_4)(x_2 + x_3)\}.$$

□

Section 4.2

Exercise (4.2.2). List the elements of S_3 as 1, (12), (23), (13), (123), (132) and label these with the integers 1, 2, 3, 4, 5, 6 respectively. Exhibit the image of each element of S_3 under the left regular representation of S_3 into S_6 .

Solution: Note that $S_3 = \langle (12), (23) \rangle$. First let's compute the image of (12) under the left regular representation of S_3 ,

$$(12)1 = (12) = \sigma_{(12)}(1) = 2$$

$$(12)(12) = 1 = \sigma_{(12)}(2) = 1$$

$$(12)(23) = (132) = \sigma_{(12)}(3) = 6$$

$$(12)(13) = (123) = \sigma_{(12)}(4) = 5$$

$$(12)(123) = (13) = \sigma_{(12)}(5) = 4$$

$$(12)(132) = (23) = \sigma_{(12)}(6) = 3.$$

Thus $\sigma_{(12)} = (12)(45)(36)$. Computing the image of (23) under the left regular representation of S_3 ,

$$(23)1 = (23) = \sigma_{(23)}(1) = 3$$

$$(23)(12) = (123) = \sigma_{(23)}(2) = 5$$

$$(23)(23) = 1 = \sigma_{(23)}(3) = 1$$

$$(23)(13) = (132) = \sigma_{(23)}(4) = 6$$

$$(23)(123) = (12) = \sigma_{(23)}(5) = 2$$

$$(23)(132) = (13) = \sigma_{(23)}(6) = 4.$$

Thus $\sigma_{(23)} = (13)(25)(46)$. Since the permutation representation of a group is a homomorphism we know that the generators have the following property, $\sigma_{S_3} = \sigma_{\langle (12), (23) \rangle} = \langle (12)(45)(36), (13)(25)(46) \rangle$. $\square$

Exercise (4.2.8). Prove that if H has finite index n then there is a normal subgroup K of G with $K \leq H$ and $|G : K| \leq n!$.

Proof: Suppose that H has finite index n in G or in other words $|G : H| = |G/H| = n$. Let G act by left multiplication on the set G/H . Now let π_H be the associated permutation

representation of this action i.e. there exists some subgroup $S_{G/H}$ of S_n isomorphic to G/H so $\pi_H : G \rightarrow S_{G/H}$. By Theorem 3(3) we get that $\ker \pi_H$ is $\cap_{x \in G} xHx^{-1}$ and that it is the largest normal subgroup of G contained in H . Quickly recalling the argument in the text, by definition and some algebra we know that

$$\begin{aligned} \ker \pi_H &= \{g \in G \mid g \cdot xH = xH \text{ for all } x \in G\}, \\ &= \{g \in G \mid x^{-1}gxH = H \text{ for all } x \in G\}, \\ &= \{g \in G \mid x^{-1}gx \in H \text{ for all } x \in G\}, \\ &= \{g \in G \mid g \in xHx^{-1} \text{ for all } x \in G\}, \\ &= \cap_{x \in G} xHx^{-1}. \end{aligned}$$

Suppose $N \trianglelefteq G$ with $N \leq H$. Therefore we know that $xNx^{-1} \leq xHx^{-1}$ for all $x \in G$. Then it follows that $N \leq \cap_{x \in G} xHx^{-1} = \ker \pi_H$. By the First Isomorphism Theorem we know that $G/\ker \pi_H \cong \pi_H(G) \leq S_{G/H} \leq S_n$. Thus $|G : \ker \pi_H| \leq n!$. $\square$

Exercise (4.2.10). Prove that every non-abelian group of order 6 has a nonnormal subgroup of order 2. Use this to classify groups of order 6. [Produce an injective homomorphism into S_3]

Proof: Suppose G is a non-abelian group, with $|G| = 6$. Note that $2|6$ and $3|6$ therefore by Cauchy's Theorem, there exists $a, b \in G$ such that $|a| = 2$ and $|b| = 3$. Consider $\langle a \rangle$, and note that since $|a| = 2$ we know that $\langle a \rangle = \{1, a\}$. Suppose for the sake of contradiction that $\langle a \rangle$ is normal in G , then for all $g \in G$ it follows that $gag^{-1} = a$, thus $a \in Z(G)$. Note that $|Z(G)| \neq 6$ as it would imply that G is abelian. Since $1, a \in Z(G)$, by Lagrange's Theorem $|G/Z(G)| = 2$ or 3 . In either case $G/Z(G)$ will be cyclic and therefore G must be abelian (Exercise 3.1.36). Thus a contradiction and therefore $\langle a \rangle$ must be nonnormal.

Consider the cosets $G/\langle a \rangle$ and note that, $|G : \langle a \rangle| = 3$. Let G act by left multiplication on $G/\langle a \rangle$. Now let $\pi_{\langle a \rangle}$ be the associated permutation representation of this action i.e. there exists some subset $S_{G/\langle a \rangle}$ of S_3 isomorphic to $G/\langle a \rangle$, so we have that

$\pi_{\langle a \rangle} : G \rightarrow S_{G/\langle a \rangle}$. Note that the $\ker \pi_{\langle a \rangle}$ must be normal in G and contained in $\langle a \rangle$, thus $\ker \pi_{\langle a \rangle} = 1$ and the map from $\pi_{\langle a \rangle} : G \rightarrow S_{G/\langle a \rangle}$ is injective. Therefore since $|G| = 6$, it follows that $G \cong S_3$.

Now suppose G were abelian. Since $2|6$ and $3|6$ by Cauchy's Theorem, there exists $a, b \in G$ such that $|a| = 2$ and $|b| = 3$. Then clearly it follows that $(ab)^6 = (a^2)^3(b^3)^2 = 1$ and since 2 and 3 are coprime and 6 is the least common multiple, so $|ab| = 6$. Thus $G = \langle ab \rangle$ and therefore $G \cong \mathbb{Z}_6$. $\square$

Section 4.3

Exercise (4.3.4). Prove that if $S \subseteq G$ and $g \in G$ then $gN_g(S)g^{-1} = N_G(gSg^{-1})$ and $gC_G(S)g^{-1} = C_G(gSg^{-1})$.

Proof: Suppose $S \subseteq G$ and fix some $g_x \in G$. Let $g \in g_x N_g(S) g_x^{-1}$ and consider $g(g_x S g_x^{-1}) g^{-1}$.

Note there exists some $g_s \in N_g(S)$ such that $g = g_x g_s g_x^{-1}$. By substitution we get that,

$$g(g_x S g_x^{-1}) g^{-1} = g_x g_s g_x^{-1} (g_x S g_x^{-1}) g_x g_s^{-1} g_x^{-1} = g_x (g_s S g_s^{-1}) g_x^{-1} = g_x S g_x^{-1}.$$

Therefore $g \in N_G(g_x S g_x^{-1})$.

Let $g \in N_g(g_x S g_x^{-1})$, and note that

$$(g_x^{-1} g g_x) S (g_x^{-1} g^{-1} g_x) = g_x^{-1} (g(g_x S g_x^{-1}) g^{-1}) g_x = g_x^{-1} (g_x S g_x^{-1}) g_x = S.$$

Therefore $g_x^{-1} g g_x \in N_g(S)$ so it follows that $g \in g_x N_g(S) g_x^{-1}$. Thus $g_x N_g(S) g_x^{-1} = N_G(g_x S g_x^{-1})$.

Let $g \in g_x C_G(S) g_x^{-1}$, and consider $g(g_x S g_x^{-1}) g$. Note that there exists some g_c such that $g = g_x g_c g_x^{-1}$. By substitution we get that,

$$g(g_x S g_x^{-1}) g = g_x g_c g_x^{-1} (g_x S g_x^{-1}) g_x g_c^{-1} g_x^{-1} = g_x (g_c S g_c^{-1}) g_x^{-1} = g_x (C_G(S)) g_x^{-1}.$$

Note that the second to last equality preserves S since $g_c \in C_G(S)$. Thus $g \in C_G(g_x S g_x^{-1})$. Let $g \in C_G(g_x S g_x^{-1})$ and note that,

$$(g_x^{-1} g g_x) S (g_x^{-1} g^{-1} g_x) = g_x^{-1} (g (g_x S g_x^{-1}) g^{-1}) g_x = g_x^{-1} (g_x S g_x^{-1}) g_x = S.$$

Again here $g (g_x S g_x^{-1}) g^{-1} = g_x S g_x^{-1}$ preserves $g_x S g_x^{-1}$ since $g \in C_G(g_x S g_x^{-1})$. Thus $g_x^{-1} g g_x \in C_g(S)$ so it follows that $g \in g_x C_g(S) g_x^{-1}$. Thus $g_x C_g(S) g_x^{-1} = C_G(g_x S g_x^{-1})$

□

Exercise (4.3.5). If the center of G is of index n , prove that every conjugacy class has at most n elements.

Proof: Suppose that $|G/Z(G)| = n$. Suppose $C_G(g)$ is a conjugacy class of G . By definition we know that $C_G(g) = O(g)$ where $O(g)$ is the orbit of g under conjugation by G . By the Orbit Stabilizer Theorem we know that $|C_G(g)| = |O(g)| = |G/G_g|$. Note that under conjugation $G_g = C_G(\{g\})$. Recall that $Z(G) = \bigcap_{g \in G} C_G(\{g\})$ and therefore $Z(G) \leq C_G(\{g\})$ for all $g \in G$. Thus by Lagrange's Theorem $|C_G(g)| = |G/C_G(\{g\})| \leq |G/Z(G)| = n$. □

Exercise (4.3.6). Assume G is a non-abelian group of order 15. Prove that $Z(G) = 1$. Use the fact that $\langle g \rangle \leq C_G(g)$ for all $g \in G$ to show that there is at most one possible class equation for G .

Proof: Suppose G is a non-abelian group of order 15. By Lagrange's Theorem we know that if G is non-abelian, then $|Z(G)| = 1, 3$ or 5 . If $|Z(G)| = 3$ then by Lagrange's Theorem $|G/Z(G)| = 5$ and therefore cyclic. By Exercise 3.1.36 it would then follow that G is cyclic and therefore abelian, a contradiction. Similarly if $|Z(G)| = 5$ then by Lagrange's Theorem $|G/Z(G)| = 3$ and therefore cyclic. Again it follows that G is cyclic and abelian, a contradiction. Therefore $Z(G) = 1$.

Suppose that $\langle g \rangle \leq C_G(g)$ for all $g \in G$. Note that $3|15$ and $5|15$ therefore by Cauchy's Theorem there exists an $a, b \in G$ such that $|a| = 3$ and $|b| = 5$. Since

$\langle a \rangle \leq C_G(a) \leq G$ and $a \notin Z(G)$ with $|G : \langle a \rangle| = 5$, we get that $C_G(a) = \langle a \rangle$. Similarly since $\langle b \rangle \leq C_G(b) \leq G$ and $b \notin Z(G)$ with $|G : \langle b \rangle| = 3$, we get that $C_G(b) = \langle b \rangle$. Applying the class equation we know that,

$$|G| = |Z(G)| + \sum_{i=1}^r |G : C_G(g_i)|,$$

For some conjugacy class representatives $g_1, g_2, \dots, g_r$. By substitution of $|G : C_G(a)|$, $|G : C_G(b)|$ and $|Z(G)|$ we get,

$$|G| = 1 + 3 + 5 + \sum_{i=1}^{r-2} |G : C_G(g_i)|.$$

Note that each summand in $\sum_{i=1}^{r-2} |G : C_G(g_i)|$ must be a divisor of $|G| = 15$ and total to 6. It cannot be $5 + 1$ since that would imply there exists a $|C_G(g_i)| = 15$ which would have $g_i \in Z(G)$. Therefore the final class equation looks like this,

$$|G| = 1 + 3 + 5 + (3 + 3).$$

□

Exercise (4.3.8). Prove that $Z(S_n) = 1$ for all $n \geq 3$.

Proof: Suppose S_n where $n \geq 3$ and suppose for the sake of contradiction that $|Z(S_n)| > 1$.

Let $\sigma \in S_n$ be a nontrivial permutation which fixes a single element $i \in [n]$, and has an arbitrary cycle decomposition $(a_1 a_2 \dots a_k)(b_1 b_2 \dots b_{k_2}) \dots$. Let $\tau \in Z(S_n)$ and note that by Proposition 10 we know that $\tau \sigma \tau^{-1} = (\tau(a_1) \tau(a_2) \dots \tau(a_k))(\tau(b_1) \tau(b_2) \dots \tau(b_{k_2})) \dots$. Since $\tau \in Z(S_n)$ it must also fix i . Let φ be of the same cycle type as σ but with the property that it doesn't fix i . Note $\varphi \in S_n$ since $n \geq 3$. Thus $\tau \varphi \tau^{-1} = (\tau(c_1) \tau(c_2) \dots \tau(c_k))(\tau(d_1) \tau(d_2) \dots \tau(d_{k_2})) \dots$. Note φ cannot equal $\tau \varphi \tau^{-1}$ since τ fixes i and φ doesn't. Thus a contradiction, and therefore $|Z(S_n)| = 1$. □

Exercise (4.3.12). Find a representative for each conjugacy class of elements of order 4 in S_8 and in S_{12} .

Solution: Recall that the number of conjugacy classes of S_n equals the number of partitions of n . Also recall that the order of a permutation is the least common multiple of the lengths of its cycle decomposition. Note that the partitions of 8 containing parts of size 2 and or just 4 are,

Partition of 8 with lcm(4)	Representative of Conjugacy Class
4, 4	(1 2 3 4)(5 6 7 8)
4, 2, 2	(1 2 3 4)(5 6)(7 8)
4, 2, 1, 1	(1 2 3 4)(5 6)
4, 1, 1, 1, 1	(1 2 3 4)

Similarly we can apply this method to solve S_{12} ,

Partition of 12 with lcm(4)	Representative of Conjugacy Class
4, 4, 4	(1 2 3 4)(5 6 7 8)(9 10 11 12)
4, 4, 2, 2	(1 2 3 4)(5 6 7 8)(9 10)(11 12)
4, 4, 2, 1, 1	(1 2 3 4)(5 6 7 8)(9 10)
4, 4, 1, 1, 1, 1	(1 2 3 4)(5 6 7 8)
4, 2, 2, 2, 2	(1 2 3 4)(5 6)(7 8)(9 10)(11 12)
4, 2, 2, 2, 1, 1	(1 2 3 4)(5 6)(7 8)(9 10)
4, 2, 2, 1, 1, 1, 1	(1 2 3 4)(5 6)(7 8)
4, 2, 1, 1, 1, 1, 1, 1	(1 2 3 4)(5 6)
4, 1, 1, 1, 1, 1, 1, 1, 1	(1 2 3 4)

□

Exercise (4.3.17). Let A be a nonempty set and let X be any subset of S_A . Let

$$F(X) = \{a \in A \mid \sigma(a) = a \text{ for all } \sigma \in X\}$$

Let $M(X) = A - F(X)$ be the elements which are moved by some element of X . Let $D = \{\sigma \in S_A \mid |M(\sigma)| < \infty\}$. Prove that D is a normal subgroup of S_A .

Proof: Consider the problem statement and suppose $D = \{\sigma \in S_A \mid |M(\sigma)| < \infty\}$. Let $\sigma \in D$ and $\tau \in S_A$. Note that by definition σ is a finite permutation and therefore has a some finite cycle permutation representation, $(a_1 a_2 \dots a_k)(b_1 b_2 \dots b_{k_2}) \dots$. Note that by

Proposition 10 we know that $\tau\sigma\tau^{-1} = (\tau(a_1)\tau(a_2)\dots\tau(a_k))(\tau(b_1)\tau(b_2)\dots\tau(b_{k_2}))\dots$. Since $\tau\sigma\tau^{-1}$ has a finite cycle permutation representation it must also have finite order, and it fixes all the elements that σ fixes and it moves every element that σ moves. Therefore $\tau\sigma\tau^{-1} \in D$ and thus $D \trianglelefteq S_A$. $\square$